UNSA · SYSTEMS ENGINEERING

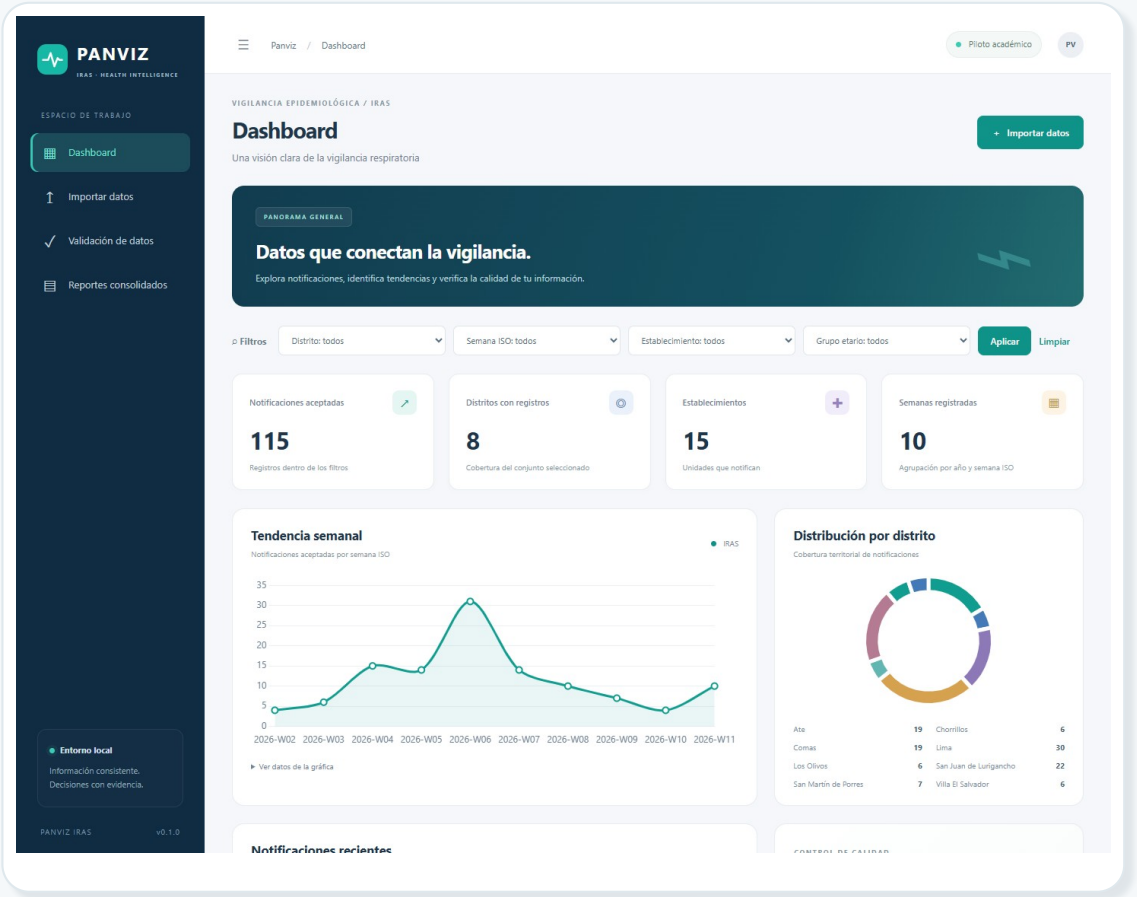
RESEARCH PROJECT

PANVIZ IRAS

Development and verification for data quality and software traceability

Software Quality · November 10, 2026

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ACADEMIC PILOT · SYNTHETIC DATA

PROBLEM

Convincing charts do not guarantee reliable data

The challenge starts with the CSV, before any dashboard.

01

Missing fields

Records missing a district or any required field must be excluded.

02

Duplicates

Repeated IDs may inflate counts without proper safeguards.

03

Invalid dates and categories

Impossible values and unsupported categories distort summaries.

No validation rules → **unreliable totals** → **unverifiable decisions**

RESEARCH

What did the study seek to demonstrate?

Research question and technical objective of the pilot.

RESEARCH QUESTION

How can we reject input errors, reconcile aggregated results, and maintain traceability across requirements and changes?

01 Validation

Testable rules and a CSV data contract

02 Persistence

SQLite and transactional confirmation

03 Verification

Automated tests and traceability

Scope: local technical case study; no real patients or clinical evaluation.

PANVIZ IRAS: four connected modules

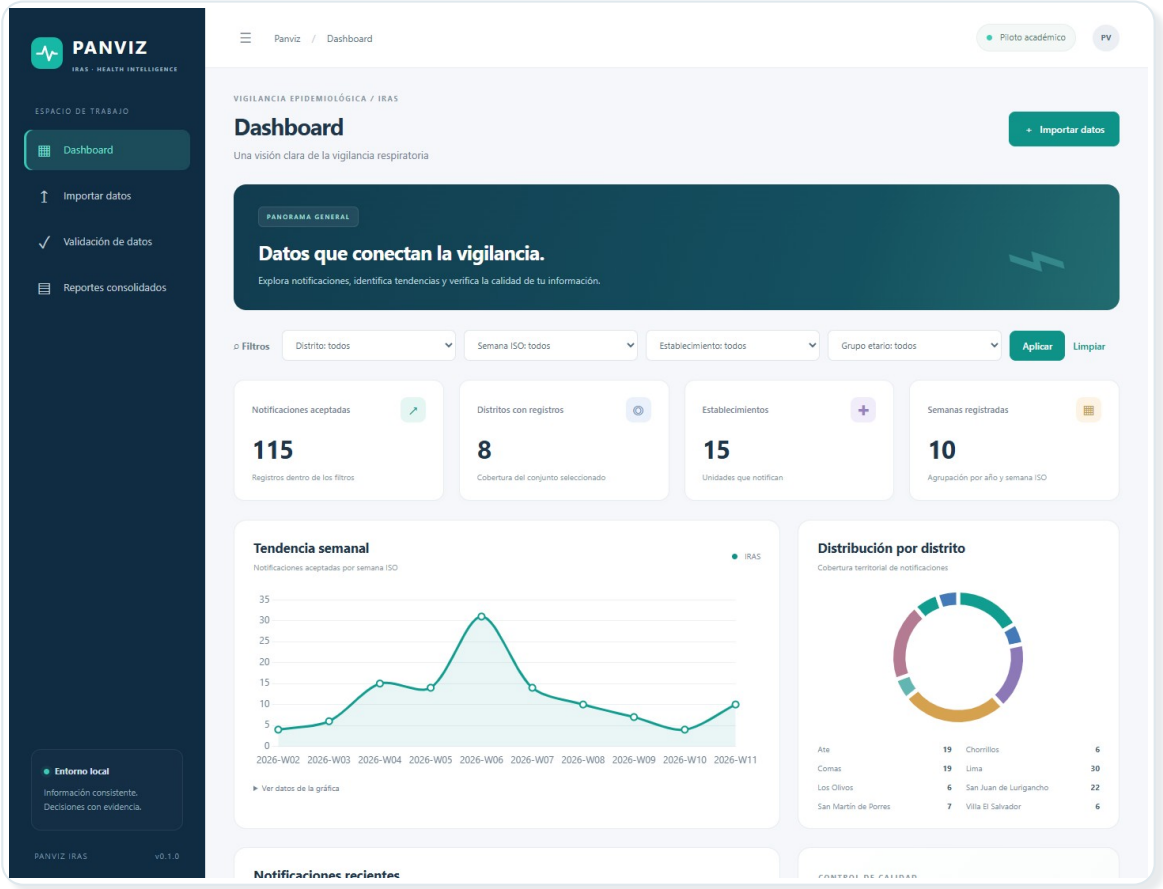
Local web app for managing synthetic acute respiratory infection reports.

01 Dashboard
Metrics, weekly trends, and districts

02 Import
CSV upload, preview, and confirmation

03 Validation
Rejection reason and source row

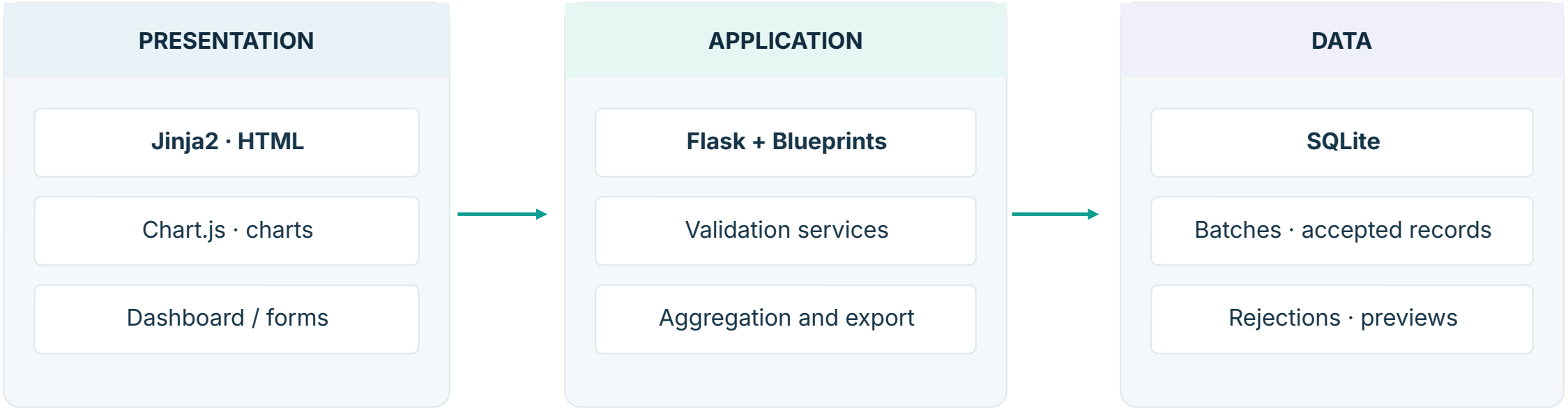
04 Reports
Grouped summaries and reconciled exports



TECHNICAL DESIGN

Modular architecture with clear responsibilities

Flask orchestrates; SQLite is the source of truth.



Isolated pytest tests + temporary SQLite databases

DATA QUALITY

An explicit, testable CSV data contract

R01 and R02: six mandatory fields and validation before storage.

01	id
02	fecha_notificacion
03	establecimiento
04	distrito
05	grupo_etario
06	evento

Accepted input

UTF-8 · optional BOM · comma or ; · 2 MB · 10,000 rows

Specific rules

Real dates · age-group catalog · IRAS only

FUNCTIONAL FLOW

From CSV input to auditable figures

Confirmation revalidates and commits data in one transaction.

1 Input CSV

Six fields

2 Preview

Row-level rules

3 Confirmation

Revalidation

4 SQLite

Accepted and rejected

Aggregation

ISO week, district, facility, age group

Reconciliation

Sum of groups = filtered total

Export

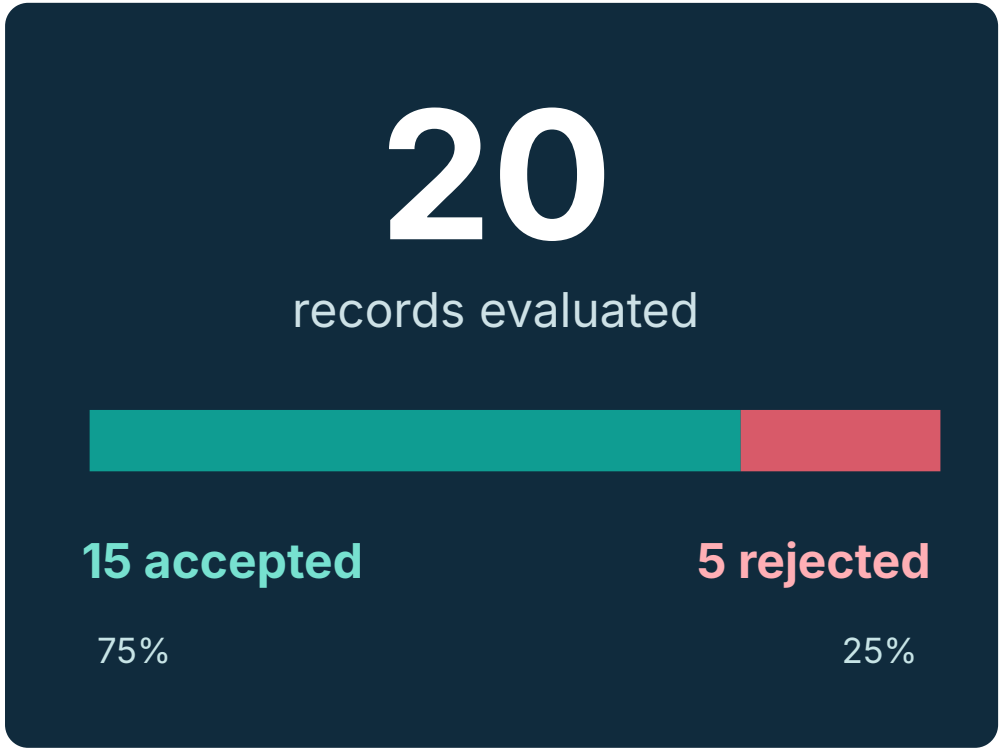
Summary / rejection CSV

Rejection reports retain the reason and original row number.

RESULT 01

The baseline dataset flagged all five expected errors

Expected outcomes defined before test execution.



Documented rejection reasons

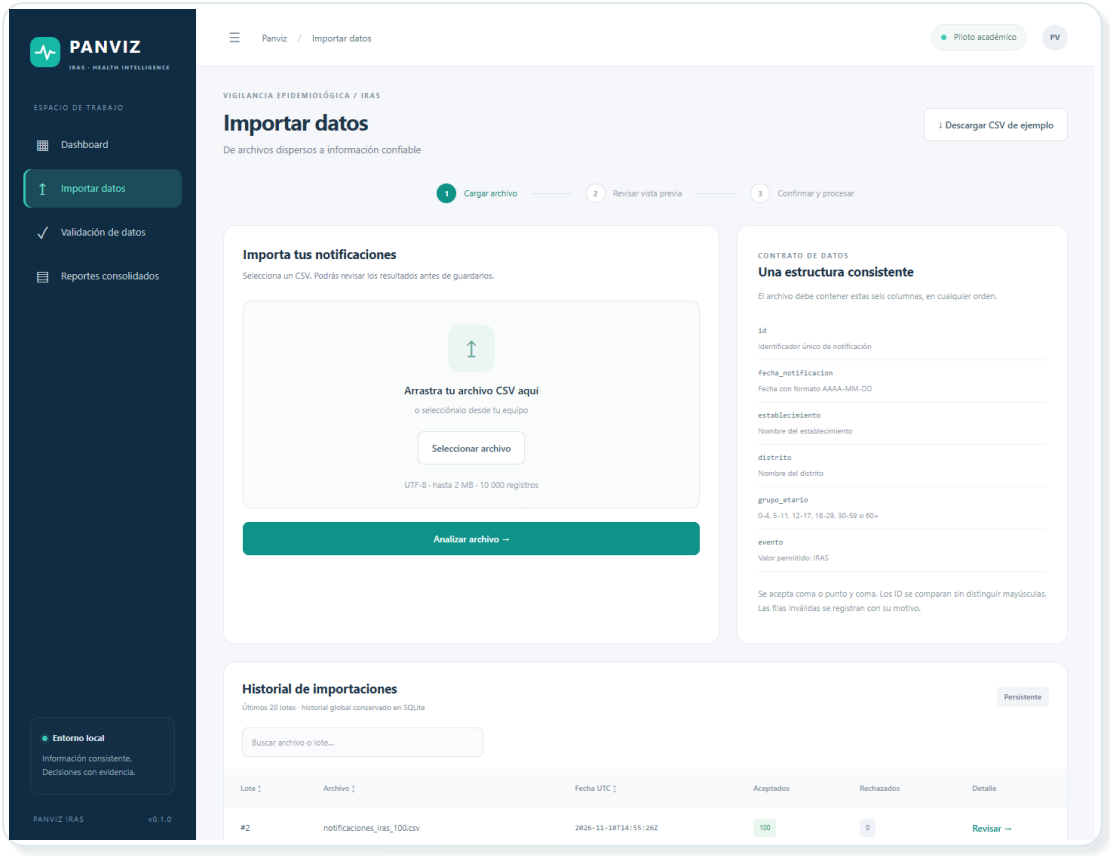
- Duplicate notification ID
- Missing mandatory district
- Impossible date
- Age group outside the catalog
- Event other than IRAS

5/5 errors detected · reconciliation gap = 0

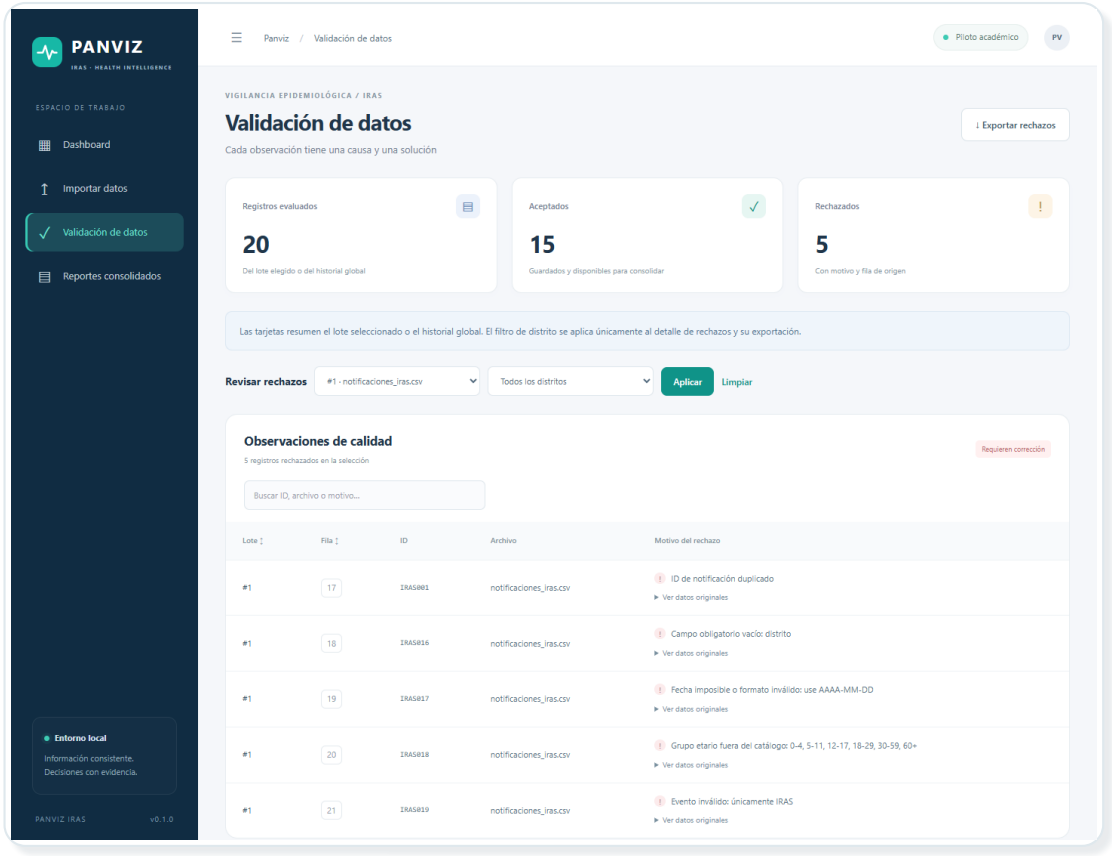
VISUAL EVIDENCE

The interface makes rules and results visible

Actual prototype screenshots from the research report.



Import: visible data contract and CSV upload



Validation: accepted, rejected, and reasons

CR01: district filter with regression tests

The same filters apply to summaries and exports.

Ate · baseline dataset

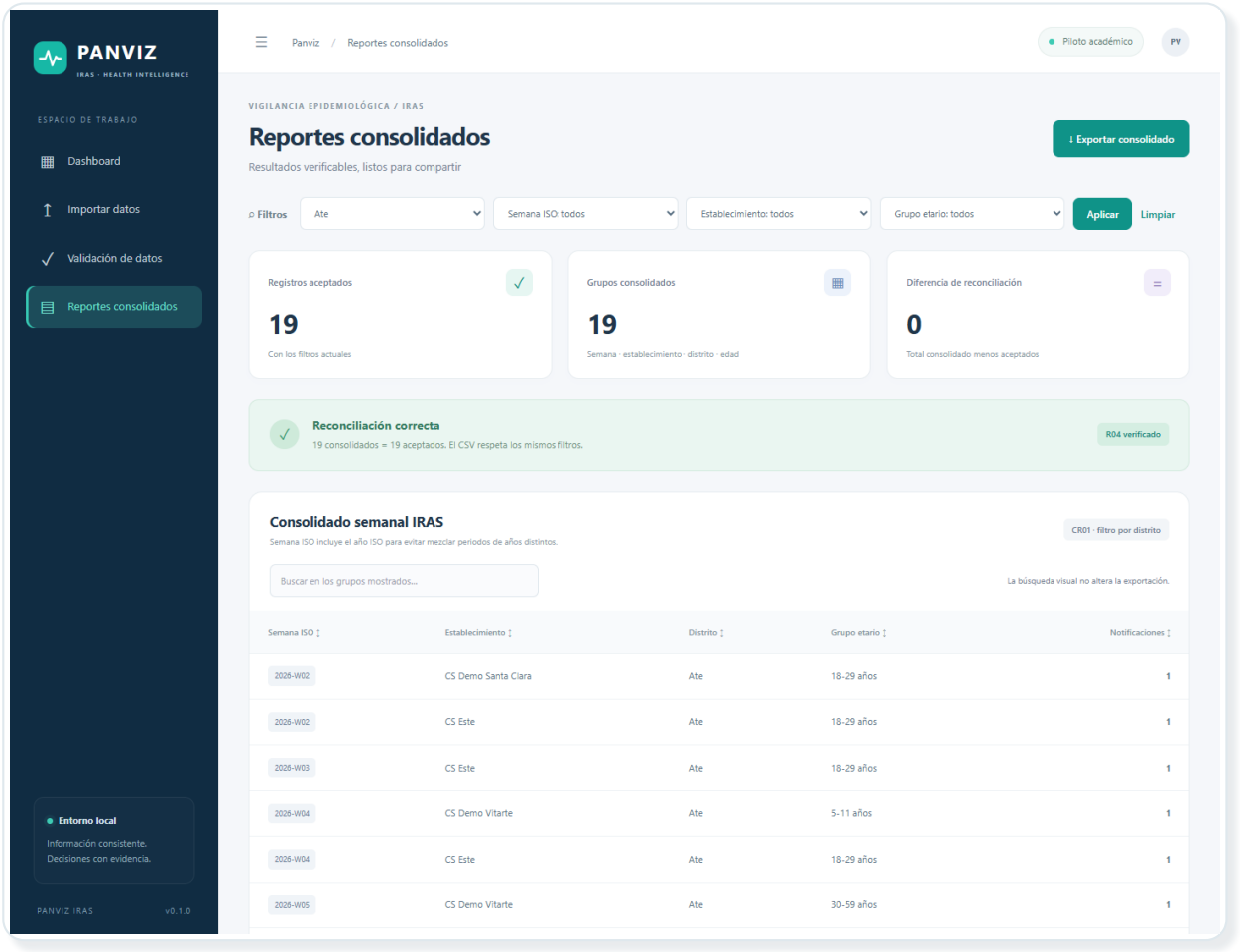
5 accepted records

Ate · expanded dataset

19 accepted records

Reconciliation gap

0 database unchanged



AUTOMATED VERIFICATION

51 tests passed in the evaluated run

Success in tested scenarios, not a guarantee of zero defects.

51

OF 51

0 failed

1	**Validation**	Fields, duplicates, formats, and limits
2	**Integration**	Routes, templates, and confirmation
3	**Data**	Temporary SQLite and transactions
4	**Results**	Aggregates, filters, and exports
5	**Traceability**	Existing traceability-matrix references

SYNTHETIC DATA EXPANSION

From 15 to 115 persisted notifications

100 valid rows were added without replacing the baseline.

15

initial accepted records

+

+100

additional accepted rows

=

115

final accepted records

8 districts · 10 ISO weeks · 6 age groups · 12 demo facilities

The 20-row test oracle remains separate from the expanded demo.

District distribution: synthetic data

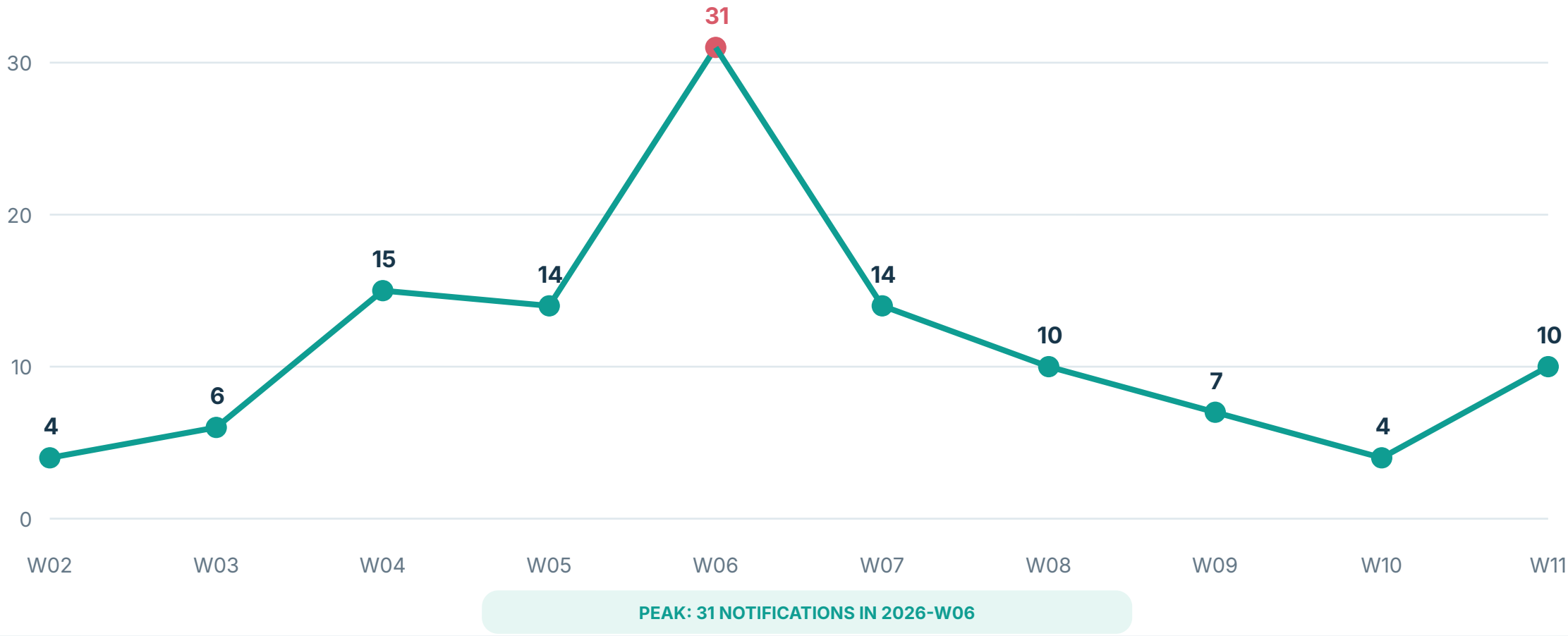
Variation reflects synthetic data generation, not official statistics.



TOTAL = 115

Weekly trend: the peak occurs in W06

Ten ISO weeks in 2026, totaling 115 synthetic records.



TECHNICAL FINDING

Uniform charts did not indicate an algorithm defect

Inspecting SQLite data explained the apparent anomaly.

1

Observation

Three districts had 5 records each; five weeks had 3 each.

2

Diagnosis

The dashboard queried 15 stored rows; the 100-row CSV had not been imported.

3

Solution

Validate and import the expanded dataset, then reconcile again; keep the aggregation logic.

Generated file ≠ imported file ≠ dashboard data

QUALITY GOVERNANCE

Functional traceability and selected CMMI practices

Documented practices do not prove organizational maturity.

R01	CSV import
R02	Validation / rejections
R03	ISO-week aggregation
R04	Export / reconciliation
R05	Traceability matrix
CR01	District filter

18 reference process areas

7 Level 2 + 11 Level 3 (CMMI-DEV v1.3)

Human evidence still pending

Independent review, user validation, training, and formal sign-off.

Code and test links were checked; their adequacy still requires human review.

CONCLUSIONS

What was demonstrated, and what remains?

Observed technical results and responsible next steps.

Observable quality

15 accepted / 5 rejected in baseline

Reproducible verification

51 tests passed in the reported run

Reconciled results

115 final acceptances and zero reconciliation gap

Next steps

User validation · independent review · performance / accessibility · security controls

Scope limit: synthetic data; no clinical effectiveness, representativeness, or CMMI certification.

PANVIZ IRAS

Thank you

Questions and discussion

SOURCES AND EVIDENCE

PANVIZ IRAS research report (2026) · Figures and app screenshots

Code and documentation: github.com/AnthoLogy63/PANVIZ_IRAS-System

Report references: WHO · SEI/CMMI v1.3 · Flask · SQLite · pytest · Python · WCAG 2.2